



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

July 7, 2016

Name, Surname:

Student ID:

Row:

Column:

Time: 2 hour 30 minutes

Prof. Marcello Restelli

Maximum Mark: 31

- The following exam is composed of **10 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used only in the large squares present on each page. Any solution provided outside these spaces will not be considered for the final mark.
- During this exam you are **not allowed** to use electronic devices like laptops, smartphones, tablets and similar. As well, you are not allowed to bring with you any kind of note, book, written scheme or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Ex. 9	Ex. 10	Tot.
/ 5	/ 5	/ 5	/ 2	/ 2	/ 2	/ 2	/ 2	/ 3	/ 3	/ 31

Exercise 1 (5 points)

Describe the supervised learning technique denominated **Support Vector Machines** for classification problems.

Exercise 2 (5 points)

Describe the supervised learning technique denominated **logistic regression** for classification problems.

Exercise 3 (5 points)

Describe the differences existing between the **Montecarlo** and the **Temporal Difference** methods in the model-free estimation of a value function for a given policy.

Exercise 4 (2 point)

Comment the following statements about adding new features to a generic model:

1. The addition of new features requires a longer time for the training of the model;
2. It is always a good idea to add some feature in a classification problem, since this may increase the chance to consider a new feature space where it is possible to linearly separate the classes;
3. You just need to add new features to improve your learner capabilities;
4. You need to know the explicit formula for computing a new feature if you want to include it in your model.

1. TRUE/FALSE: This is true if you are considering a parametric model, since the training complexity might increase. If we are resorting to a non-parametric model, which does not consider a training phase, this is not true anymore;
2. FALSE: The addition of new features, even if increases the generalization capabilities of the model, might have a negative effect since we should always balance between the bias and variance of the considered model. If we have an unnecessary complex model, we might overfit over the training samples;
3. FALSE: The reasons are the same of point 2;
4. FALSE: A counterexample is the use of the kernel trick, with which we are able to consider even infinitely large feature space without even knowing the explicit features we consider.

Exercise 5 (2 point)

Consider the following statement regarding **Principal Component Analysis** (PCA) and tell if they are true or false. Provide motivation for your answers.

1. PCA might get stuck into local optima, thus trying multiple random initializations might help;
2. Even if all the input features have similar scales, we should still perform mean normalization (so that each feature has zero mean) before running PCA;
3. Given the scores $\tilde{x}_i, \forall i$ and the loadings matrix W , there is no way to reconstruct the original samples $x_i, \forall i$;
4. PCA can be used either for data compression, data visualization or feature extraction.

1. FALSE: There is no source of stochasticity in the process of performing PCA. Thus the use of multiple initialization would not produce different results;
2. TRUE: The process considers the case in which the points are centered in the origin (have zero mean). If one of the component has an average value far different from zero, we might bias the direction of the first eigen-vector towards this dimension;
3. FALSE/TRUE: If we consider a zero mean dataset, we can exploit the orthogonality of the matrix W to reconstruct the original values ($x_i = \tilde{x}_i W$), otherwise we also need the values of the expected values of each dimension μ_i ($x_i = \tilde{x}_i W + \mu_i$);
4. TRUE: PCA naturally tries to minimize the euclidean error between the reconstructed and the original data, thus it can be used for data compression (considering the first k columns of the loadings matrix W). If we consider only the first 2–3 principal components, we are able to visualize the data, even if their original dimensionality did not allow to represent them. At last, PCA can be considered also as an unsupervised techniques for feature extraction, where we consider the first k principal components as new features.

Exercise 6 (2 point)

Consider the case in which you have an estimated state/action value of $Q(s, a) = 3$ you perform action a , gain a reward of $R_t = 1$ and reach state s' . In s' you could perform only actions a'_1 and a'_2 ($Q(s', a'_1) = 1$ and $Q(s', a'_2) = 2$). Moreover, if you would use the current policy π you would choose action a'_1 in state s' .

Consider a discount factor $\gamma = 0.5$ and a learning rate $\alpha = 0.5$. Tell if the following values are consistent with the use of SARSA and/or Q-learning algorithms after the update:

1. $Q(s, a) = 2.75$
2. $Q(s, a) = 2.25$
3. $Q(s, a) = 3$
4. $Q(s, a) = -2.5$

Motivate the answers you provided.

The update rules in this case of SARSA and Q-learning are, respectively:

$$\begin{aligned} Q(s, a) &\leftarrow Q(s, a) + \alpha(R_t + \gamma Q(s', a'_1) - Q(s, a)) \\ &= 3 + 0.5(1 + 0.5 \cdot 1 - 3) = 2.25 \\ Q(s, a) &\leftarrow Q(s, a) + \alpha(R_t + \gamma \max_{a'} Q(s', a') - Q(s, a)) \\ &= 3 + 0.5(1 + 0.5 \cdot 2 - 3) = 2.5 \end{aligned}$$

Thus:

1. Inconsistent with both;
2. Consistent with SARSA;
3. Inconsistent with both;
4. Inconsistent with both.

Exercise 7 (2 point)

Which of the following is **not** a linear function in x :

1. $f(x) = \text{logit}(P(y = 1|x))$, where $P(y = 1|x) = \frac{1}{1+\exp\{w^T x + b\}}$, $w, x \in \mathbb{R}^d$ and $b \in \mathbb{R}$
2. $g(x) = \frac{x-1}{x+1}$
3. $h(x) = \frac{x^2-1}{x+1}$
4. $i(x) = x \frac{\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2} + \log(\pi)$, with $\mu, \sigma \in \mathbb{R}$

Motivate your answers.

Hint: $\text{logit}(x) = \ln\left(\frac{x}{1-x}\right)$.

1. Linear: $f(x) = -w^T x - b$;
2. Not linear: $\lim_{x \rightarrow \infty} g(x) = 1$;
3. Linear/Not linear: If we exclude $x = -1$ it can be simplified to $h(x) = x - 1$, if we consider also $x = -1$ we have a discontinuity there;
4. Linear: if $m = \frac{\mu}{\sigma^2}$ and $q = -\frac{\mu^2}{2\sigma^2} + \log(\pi)$ we have $i(x) = mx + q$.

Exercise 8 (2 point)

Tell if the following statements about **Monte Carlo** (MC) and **Temporal Difference** (TD) estimation methods are true or false and motivate your answers.

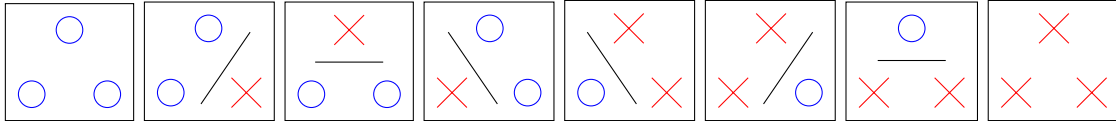
1. Given an episode of length T , you can generate T samples for the MC estimation of the value function;
2. Generally, every-visit MC estimation is better than first-visit MC estimation if you consider large amount of episodes;
3. TD cannot be used in the case there is no terminal state in original MDP;
4. MC estimation works better than TD estimation if the problem is not Markovian.

1. TRUE/FALSE: If we consider every-visit MC estimation, we will have a sample of the value function for each state we visited (thus a total of T samples). With first-visit you generate a number of samples which is at most the number of the states of the MDP you are considering;
2. FALSE: Since every-visit MC estimation provides a biased estimation of the value function for the states, it is a better idea to use first-visit MC estimation if we have a large amount of episodes;
3. FALSE: Since TD updates occurs any time a transition occurs, it does not require to wait for the terminal state for the value estimation. Thus TD can be used also in MDPs where no terminal state is present;
4. TRUE: TD intrinsically exploits the Markovian property of the MDP, since it is based on the Bellman expectation operator, while MC estimates does not consider this assumption.

Exercise 9 (3 points)

Show that the VC dimension of a linear classifier in 2D is 3.

Let us call the class of all the linear classifier in 2D $\mathcal{H}$. The proof consists in two steps: $VC(\mathcal{H}) \geq 3$ and $VC(\mathcal{H}) \leq 3$. Let us consider the first step. We need to show that it exists a set of 3 points that can be shattered by the considered hypothesis space $\mathcal{H}$. By considering a set of three non-aligned points, it is possible to show by enumeration that it is possible to shatter them with a linear classifier:



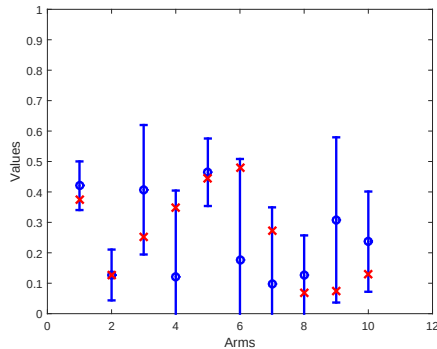
Thus, $VC(\mathcal{H}) \geq 3$. Let us consider the second step. We need to show that it does not exist a set of 4 points which can be shattered by a linear classifier. There are different cases:

- Four aligned points: if we alternate instances coming from the positive and negative classes we cannot shatter them;
- Three aligned points and a fourth on an arbitrary position: if we alternate instances coming from the positive and negative classes for the three aligned points, we cannot shatter them;
- Four points on a convex hull: if we label the points on the two diagonals with opposite classes, we cannot shatter them;
- Three points on a convex hull (triangle) and one inside the hull: if we label the three points on the triangle with a label and the last one with the other class, we cannot shatter them.

Since there does not exist a configuration where we can shatter the points, we have that $VC(\mathcal{H}) \leq 3$. $\square$

Exercise 10 (3 points)

Consider the bounds in Figure 1 (supposed to hold with probability at least δ) for a **Multi-Armed Bandit** (MAB) stochastic setting, where the bars are the Hoeffding bounds for the expected rewards, the circles are the estimated expected rewards and the crosses are the real expected rewards.



1. Which arm would a UCB1 algorithm choose for the next round?
2. Do you think that Figure 1 might be the results obtained by running UCB1 for several rounds?
3. Which arm will UCB1 converge to if $T \rightarrow \infty$?
4. Which arm will Thompson Sampling converge to if $T \rightarrow \infty$ (starting from a uniform prior)?
5. Which arm is the one which we pulled the most in Figure 1?

Figure 1: Bounds in a MAB setting. Motivate your answers.

1. a_3 , since it has the highest upper bound;
2. No, since the upper bounds should be all at similar heights;
3. a_6 , since it is the arm with the highest real expected value;
4. a_6 , since it is the arm with the highest real expected value;
5. Either a_1 or a_2 , since they are the ones with the tightest bounds.